PY101 - Introduction to Python

Week 2: Decisions and Loops

Lesson 1: Control Flow Overview: Decisions vs. Repetition

Control flow allows your program to make choices or perform repetitive actions [cite: 5]. Selection (if/else) chooses a path, while Repetition (loops) repeats code blocks [cite: 5].

Lesson 2: If/else: Syntax, Indentation, and Mental Model

Python uses indentation to group code blocks under conditions. Every `if` and `else` statement must end with a colon (`:`), and the code following them must be indented [cite: 5].

Lesson 3: Logical Operators: and/or/not

Logical operators allow for complex conditional checks [cite: 5]. Python uses **short-circuiting** to skip unnecessary checks if the final result is already determined [cite: 5].

Lesson 4: Booleans and Comparisons

Booleans are `True` or `False`. Always use == for checking equality and = for variable assignment. While `is` checks if two objects are the same in memory, == is preferred for comparing values [cite: 5].

Lesson 5: String Membership with 'in'

The `in` operator checks if a substring exists within a larger string, returning a boolean [cite: 5].

Lesson 6: Elif and Refactoring

Use `if/elif/else` chains to flatten nested decision logic. Python processes these branches from top to bottom and runs only the *first* block that evaluates to `True` [cite: 5].

Lesson 7: Decision Tables

Decision tables are a structured way to lay out conditions, ensuring logic is both **complete** and **mutually exclusive** [cite: 5].

Lesson 8: For Loops and range()

The `range()` function generates a sequence of numbers. `range(start, stop, step)` is the standard pattern, where the `stop` value is exclusive [cite: 5].

Lesson 9: The While Loop

Use a `for` loop when you know the number of iterations; use a `while` loop when you want to repeat until a specific condition changes [cite: 5]. Ensure the loop variable changes to avoid infinite loops [cite: 5].

Lesson 10: Loop Control (Break/Continue)

`break` exits a loop entirely, while `continue` skips the current iteration and jumps to the next cycle [cite: 5].

Lesson 11: Counters and Totals

A counter tracks "how many" times an action occurs; a total accumulates values. Initialize these variables *before* the loop to maintain their value across iterations [cite: 5].